

MongoDB Access Layer Architecture

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Service Layer

business logic & orchestration

- Domain logic, use cases, multi-step flows
- Orchestrates repos, validations, permissions, locks
- Exposes meaningful functions to controllers
- Enforces rules: e.g. check lock → write to Mongo

EXAMPLES

```
services/audit.service.ts
```

```
services/hdt-metadata.service.ts
```

calls

R

Repository Layer

collection access & query translation

- Encapsulates access to MongoDB collections
- Translates domain ops into concrete Mongo queries
- Handles CRUD, aggregations, indexes, lookups
- Hides DB/collection details from the service layer

EXAMPLES

```
repositories/audit.repository.ts
```

```
repositories/hdt.repository.ts
```

calls

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Mongo Infrastructure

client lifecycle & database exposure

- Creates and reuses the MongoClient (ping / reconnect)
- Exposes databases: getAuditDb(), getContentDb()
- Handles connection details, reconnect, client close
- Single entry point — never use MongoClient elsewhere

EXAMPLE

```
lib/mongo/client.ts
```

TCP

